THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — MATHS FULL SYLLABUS PAPER SET 04

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The value of 11^3 is: 1

(a) 1342 **(b)** 1320 **(c)** 1331 **(d)** 363

2. Which of the following is a perfect cube? 1

(a) 500 **(b)** 600 **(c)** 700 **(d)** 512

3. $\left(\frac{2}{5}\right)^3$ equals: 1

(a) $\frac{6}{15}$ **(b)** $\frac{8}{15}$ **(c)** $\frac{8}{125}$ **(d)** $\frac{2}{125}$

4. The mathematician who fully explained computation with the Indian ten-symbol system around 499 CE was: 1

(a) Fibonacci **(b)** Euclid **(c)** Al-Kindi **(d)** Aryabhata

5. The diagonals of a rhombus are 6 cm and 8 cm. Its side is: 1

(a) 6 cm **(b)** 4 cm **(c)** 5 cm **(d)** 7 cm

6. A rectangle that is also a rhombus is a: 1

(a) square **(b)** trapezium

(c) kite **(d)** parallelogram only

7. Is 5238 divisible by 9? 1

(a) No **(b)** Only if the number is prime

(c) Only if the last digit is even **(d)** Yes

8. If $a = 2$ and $b = 7$, then $(a + b)^2$ equals: 1

(a) 53 **(b)** 25 **(c)** 67 **(d)** 81

9. If the cost of 6 pens is ₹ 90, the cost of 10 pens is: 1

(a) ₹ 150 **(b)** ₹ 120 **(c)** ₹ 180 **(d)** ₹ 100

10. The ratio of 1 m to 40 cm is: 1

(a) 2 : 5 **(b)** 1 : 40 **(c)** 5 : 2 **(d)** 25 : 1

11. A quantity of 2000 is decreased by 18%. The new value is: 1

(a) 2360 **(b)** 2000 **(c)** 1640 **(d)** 360

12. 7.5% of 800 is: 1

(a) 50 **(b)** 70 **(c)** 60 **(d)** 120

13. The hypotenuse of a right triangle is 25 cm and one leg is 7 cm. The other leg is: 1

(a) 23 cm **(b)** 18 cm **(c)** 24 cm **(d)** 25 cm

14. A Pythagorean triplet is: 1

(a) three consecutive numbers **(b)** three numbers with $a + b = c$

(c) three numbers satisfying $a^2 + b^2 = c^2$ **(d)** three prime numbers

15. On a map, 1 cm represents 150 km. A distance of 3 cm on the map represents: 1

(a) 600 km **(b)** 225 km **(c)** 450 km **(d)** 900 km

16. In the Sierpinski triangle, the number of small shaded triangles after stage 3 is: 1

(a) 54 **(b)** 26 **(c)** 30 **(d)** 27

17. For the data 2, 2, 3, 4, 9, the best measure of a typical value unaffected by 9 is the: 1

(a) mean **(b)** median **(c)** range **(d)** sum

18. In a number pyramid, each block is the sum of the two below it. The bottom row is 3, 5, 6. The top block is: 1

(a) 21 **(b)** 24 **(c)** 19 **(d)** 14

19. Two triangles on the same base and between the same parallels have: 1

(a) different area **(b)** double area **(c)** equal area **(d)** zero area

20. Area of a triangle with sides 3, 4, 5 cm is: 1

(a) 10 cm² **(b)** 7.5 cm² **(c)** 12 cm² **(d)** 6 cm²

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Simplify $\dfrac{10^5 \times 10^{-2}}{10^2}$. 2

22. Write XLIV in decimal. 2

23. Is a number divisible by 12 if it is divisible by 3 and 4? Give a reason. 2

24. A recipe uses 2 cups of rice for 5 persons. How many cups for 15 persons? 2

25. Is the triangle with sides 5 cm, 6 cm and 8 cm right-angled? 2

26. A car takes 5 hours at 60 km/h. How long at 75 km/h? 2

27. What are outliers? Give an example. 2

28. Convert 3.5 m² to cm². 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 30–40 words each.

29. A square garden has area 1296 m². Find its side. How much fencing is needed for the boundary? 3

30. Name the quadrilateral formed by joining the midpoints of consecutive sides of a rectangle and state a reason. 3

31. Expand and simplify $(x + 3)(x + 4)$. Verify for $x = 1$. 3

32. A student scored 432 out of 600. Express as a percentage and say the fraction. 3

33. Draw the nets of a cube and a cuboid. Mention one difference. 3

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34. Find the largest product of two whole numbers with sum 41 and verify against 20 × 21. 3

35. The sides of a triangle are 15 cm, 20 cm and 25 cm. Show it is right-angled and find its area. 3

OR

Using $(n,\ \frac{n^2-1}{2},\ \frac{n^2+1}{2})$ find the triplet for $n = 5$ and verify. 3

36. A triangular field has base 22 m and height 15 m. Find its area and the cost of levelling at ₹ 12 per m². 3

OR

The area of a rhombus is 168 cm² and one diagonal is 24 cm. Find the other diagonal and the side. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. **Case study:** A dealer's annual turnover is ₹ 4.8 crore. 5

(i) Write it in standard form.

(ii) If it grows by 10^6 every year, find the turnover after 5 years.

(iii) Write the answer in standard form.

OR

(i) The diameter of a hydrogen atom is about 10^{-10} m and of a pollen grain about 10^{-5} m. How many times larger is the pollen grain? 5

(ii) Express a red blood cell of 7×10^{-6} m in mm.

38. **Case study:** A farmer borrows ₹ 50000 at 12% simple interest for 3 years. 5

(i) Find the interest.

(ii) Total to be repaid.

(iii) Monthly instalment if repaid equally over 36 months.

OR

(i) In a class, 60% of 40 students are girls. Of these, 25% have bicycles. How many girls have bicycles? 5

(ii) What percent of the class are they?

39. **Case study:** Marks of two students in 5 tests: A: 70, 72, 74, 76, 78; B: 40, 60, 74, 88, 98. 5

(i) Find both means.

(ii) Which is more consistent?

(iii) Find the ranges.

OR

(i) Four numbers have mean 12. If three of them are 8, 10 and 14, find the fourth. 5

(ii) Find the median of the four numbers.

40. (i) Find the area of an L-shaped region made of rectangles 10×4 and 6×3. 5

(ii) Suggest another method.

OR

(i) Show that the area of a rhombus is $\frac{1}{2}d_1d_2$. 5

(ii) Find the diagonal if the area is 120 cm² and the other diagonal is 15 cm.

END OF QUESTION PAPER — Total: 80 Marks

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